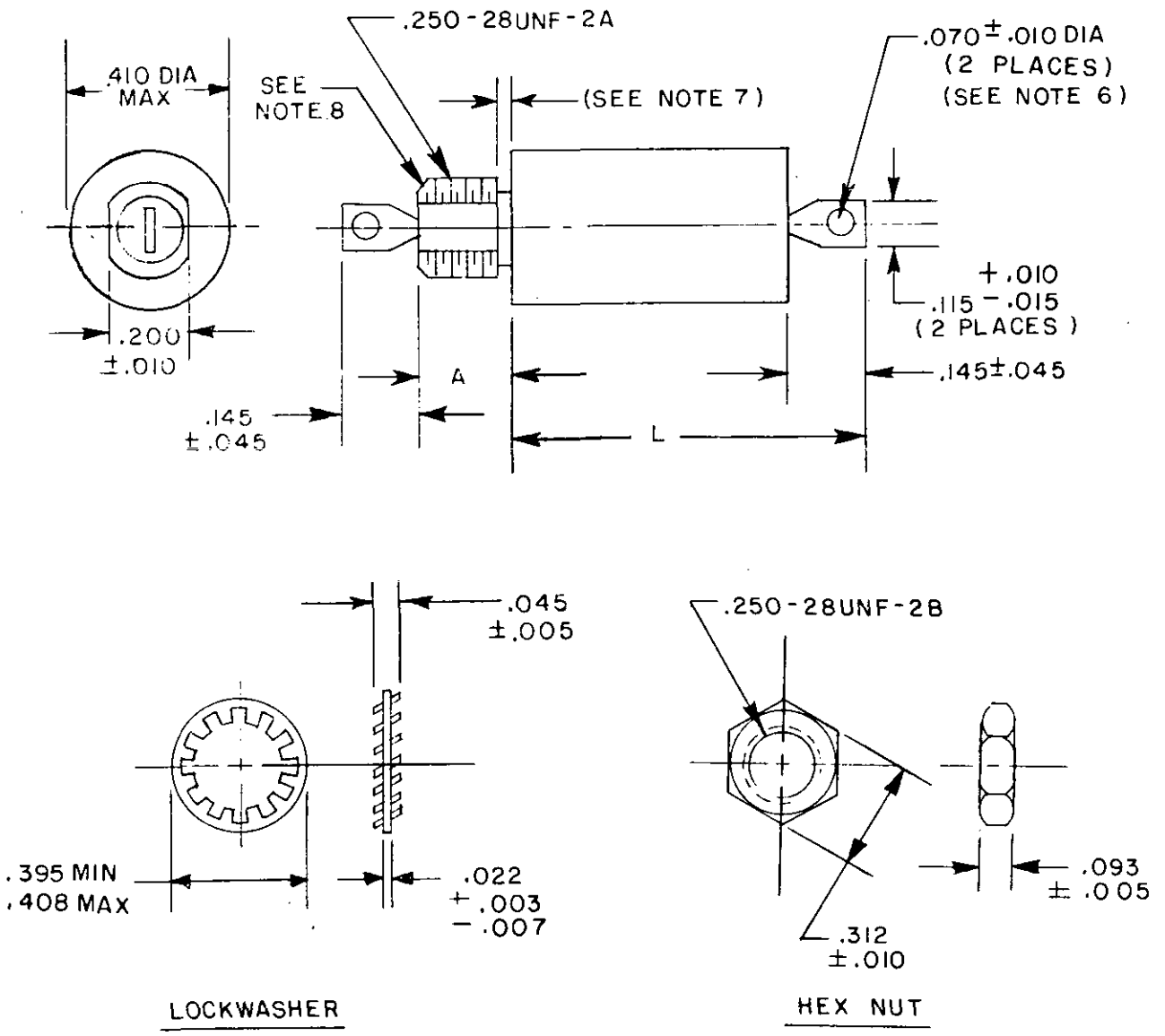
MILITARY SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE

SUPPRESSION, HERMETICALLY SEALED, STYLE FS40

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-28861.




MOUNTING HARDWARE

FIGURE 1. Case and hardware dimensions and circuit diagrams.

* U.S. GOVERNMENT PRINTING OFFICE: 1982—505-022/817

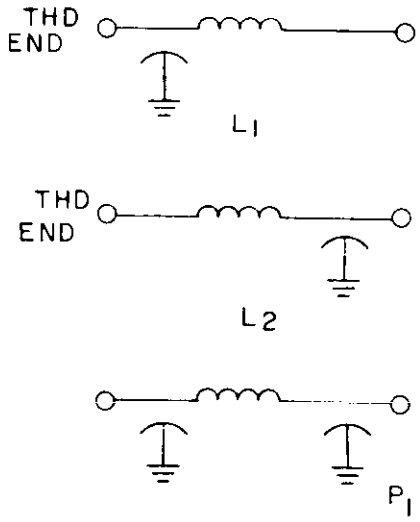


FSC 5915

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| Dash no. | A dim. | L dimension max | Dash no. | A dim | L dimension max |
|---|---|---|---|---|---|
| 001 | .187 ±.010 | .630 | 019 | .312 ±.010 | .630 |
| 002 | | .630 | 020 | | .630 |
| 003 | | .730 | 021 | | .730 |
| 004 | | .630 | 022 | | .630 |
| 005 | | .630 | 023 | | .630 |
| 006 | | .730 | 024 | | .730 |
| 007 | | .630 | 025 | | .630 |
| 008 | | .630 | 026 | | .630 |
| 009 | | .730 | 027 | | .730 |
| 010 | | .630 | 028 | | .630 |
| 011 | | .630 | 029 | | .630 |
| 012 | | .730 | 030 | | .730 |
| 013 | | .630 | 031 | | .630 |
| 014 | | .630 | 032 | | .630 |
| 015 | | .730 | 033 | | .730 |
| 016 | | .630 | 034 | | .630 |
| 017 | | .630 | 035 | | .630 |
| 018 | | .730 | 036 | | .730 |

| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .003 | .08 | .145 | 3.68 |
| .005 | .13 | .187 | 4.75 |
| .007 | .18 | .200 | 5.08 |
| .010 | .25 | .250 | 6.35 |
| .015 | .38 | .312 | 7.92 |
| .022 | .56 | .395 | 10.03 |
| .035 | .89 | .400 | 10.16 |
| .045 | 1.14 | .408 | 10.36 |
| .050 | 1.27 | .410 | 10.41 |
| .070 | 1.78 | .630 | 16.00 |
| .093 | 2.36 | .730 | 18.54 |
| .115 | 2.92 | | |




CIRCUIT DIAGRAMS

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
5. Terminal identification (non-symmetrical filters): The case shall be marked at the threaded end of the filter, with the symbol "C" or the symbol "L", as follows, or the circuit diagram shall be marked on the case.

| Circuit | Symbol |
|---|---|
| $L_1$ - - - - - - - - - | C |
| $L_2$ - - - - - - - - - | L |

6. Optional slot may be supplied, .050 ±.010 inches x .070 ±.010 inches.
7. Imperfect thread or undercut optional .050 inches maximum.
8. One imperfect thread allowed .035 inches maximum.
9. Recommended mounting torque: 44 oz-in ±4 oz-in.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.



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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 10 grams, maximum.

Case finish:

Class B filters, T only (tin-plated or tin-lead plated).

Class S filters, T or G only (tin-plated, tin-lead plated or gold plated). In accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55° to +125°C.

Rated voltage: 70 Vdc.

Rated current: See table I.

Capacitance: See table I for minimum capacitance value..

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohms, minimum.

At +125°C: 100 megohms, minimum.

Insertion loss: See table I.

Voltage drop: See table I.

DC resistance: See table I.

Seal: In accordance with MIL-F-28861. Leakage rate for class S filters shall not exceed 1 x $10^{-7}$ atm cc/sec.

Temperature rise: 25°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Product assurance level: In accordance with table I.

Part number: The part number shall be as follows:

| M28861/04 - | XXX | X | X |
|---|---|---|---|
| | Dash no. per figure 1 and table I | Case finish | Product assurance level per MIL-F-28861 and table I. |



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TABLE I. Electrical characteristics.

| Dash number | Product assurance level Class S | Class B | Circuit | Maximum current (amps) | Minimum Capacitance (μF) | Maximum voltage drop (volts) | Maximum DC resistance (ohms) | Minimum insertion loss (dB) in accordance with MIL-STD-220 1/27: At +25°C | | | | | | | | | | At -55° and +125°C | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | 15 kHz | 30 kHz | 50 kHz | 100 kHz | 150 kHz | 300 kHz | 1 MHz | 10 MHz | 100 MHz | 1 GHz | 15 kHz | 30 kHz | 50 kHz | 100 MHz | 150 MHz | 300 MHz | 1 GHz | 10 MHz | 100 MHz | 1 GHz |
| 001, 019 | X | X | L1 | 0.10 | 0.70 | 0.17 | 1.7 | 9 | 20 | 29 | 41 | 48 | 60 | 70 | 70 | 70 | 70 | 7 | 18 | 27 | 39 | 46 | 58 | 68 | 70 | 70 | 70 |
| 002, 020 | X | X | L2 | 0.10 | 0.70 | 0.17 | 1.7 | 9 | 20 | 29 | 41 | 48 | 60 | 70 | 70 | 70 | 70 | 7 | 18 | 27 | 39 | 46 | 58 | 68 | 70 | 70 | 70 |
| 003, 021 | X | X | π | 0.10 | 1.4 | 0.17 | 1.7 | 15 | 36 | 50 | 69 | 79 | 80 | 80 | 80 | 80 | 80 | 13 | 34 | 48 | 67 | 77 | 78 | 78 | 80 | 80 | 80 |
| 004, 022 | X | X | L1 | 0.30 | 0.70 | 0.23 | 0.77 | 6 | 15 | 23 | 35 | 42 | 54 | 70 | 70 | 70 | 70 | 4 | 13 | 21 | 33 | 40 | 68 | 68 | 70 | 70 | 70 |
| 005, 023 | X | X | L2 | 0.30 | 0.70 | 0.23 | 0.77 | 6 | 15 | 23 | 35 | 42 | 54 | 70 | 70 | 70 | 70 | 4 | 13 | 21 | 33 | 40 | 68 | 68 | 70 | 70 | 70 |
| 006, 024 | X | X | π | 0.30 | 1.4 | 0.23 | 0.77 | - | 29 | 44 | 62 | 73 | 80 | 80 | 80 | 80 | 80 | - | 27 | 42 | 60 | 71 | 78 | 78 | 80 | 80 | 80 |
| 007, 025 | X | X | L1 | 0.50 | 0.70 | 0.18 | 0.36 | 5 | 12 | 19 | 29 | 36 | 48 | 69 | 70 | 70 | 70 | 3 | 10 | 17 | 27 | 34 | 46 | 67 | 70 | 70 | 70 |
| 008, 026 | X | X | L2 | 0.50 | 0.70 | 0.18 | 0.36 | 5 | 12 | 19 | 29 | 36 | 48 | 69 | 70 | 70 | 70 | 3 | 10 | 17 | 27 | 34 | 46 | 67 | 70 | 70 | 70 |
| 009, 027 | X | X | π | 0.50 | 1.4 | 0.18 | 0.36 | - | 21 | 37 | 56 | 67 | 80 | 80 | 80 | 80 | 80 | - | 19 | 35 | 54 | 65 | 78 | 78 | 80 | 80 | 80 |
| 010, 028 | X | X | L1 | 1.0 | 0.70 | 0.14 | 0.14 | 5 | 11 | 15 | 21 | 26 | 36 | 55 | 70 | 70 | 70 | 3 | 9 | 13 | 19 | 24 | 34 | 53 | 70 | 70 | 70 |
| 011, 029 | X | X | L2 | 1.0 | 0.70 | 0.14 | 0.14 | 5 | 11 | 15 | 21 | 26 | 36 | 55 | 70 | 70 | 70 | 3 | 9 | 13 | 19 | 24 | 34 | 53 | 70 | 70 | 70 |
| 012, 030 | X | X | π | 1.0 | 1.4 | 0.14 | 0.14 | - | - | 20 | 46 | 57 | 75 | 80 | 80 | 80 | 80 | - | - | 18 | 44 | 55 | 73 | 78 | 80 | 80 | 80 |
| 013, 031 | X | X | L1 | 3.0 | 0.70 | 0.15 | 0.05 | 5 | 10 | 14 | 20 | 24 | 31 | 45 | 70 | 70 | 70 | 3 | 8 | 12 | 18 | 22 | 29 | 43 | 70 | 70 | 70 |
| 014, 032 | X | X | L2 | 3.0 | 0.70 | 0.15 | 0.05 | 5 | 10 | 14 | 20 | 24 | 31 | 45 | 70 | 70 | 70 | 3 | 8 | 12 | 18 | 22 | 29 | 43 | 70 | 70 | 70 |
| 015, 033 | X | X | π | 3.0 | 1.4 | 0.15 | 0.05 | - | - | - | 17 | 36 | 51 | 80 | 80 | 80 | 80 | - | - | - | 15 | 34 | 49 | 78 | 80 | 80 | 80 |
| 016, 034 | X | X | L1 | 5.0 | 0.70 | 0.075 | 0.015 | - | - | - | 14 | 17 | 24 | 36 | 60 | 70 | 70 | - | - | - | 12 | 15 | 22 | 34 | 60 | 70 | 70 |
| 017, 035 | X | X | L2 | 5.0 | 0.70 | 0.075 | 0.015 | - | - | - | 14 | 17 | 24 | 36 | 60 | 70 | 70 | - | - | - | 12 | 15 | 22 | 34 | 60 | 70 | 70 |
| 018, 036 | X | X | π | 5.0 | 1.4 | 0.075 | 0.015 | - | - | - | - | 16 | 38 | 75 | 80 | 80 | 80 | - | - | - | - | 14 | 36 | 73 | 80 | 80 | 80 |

1/ Insertion loss measurements shall be made under full-load over the frequency range of 100 kHz to 10 MHz. Insertion loss measurements above and below this frequency range shall be under no-load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.





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